INFORMATICS INSTITUTE OF TECHNOLOGY

In Collaboration with

ROBERT GORDON UNIVERSITY ABERDEEN

NUTRI SCANNER

(Personalized Nutrition Tracking and Health Risk Prediction System)

A Requirement Specification Document by

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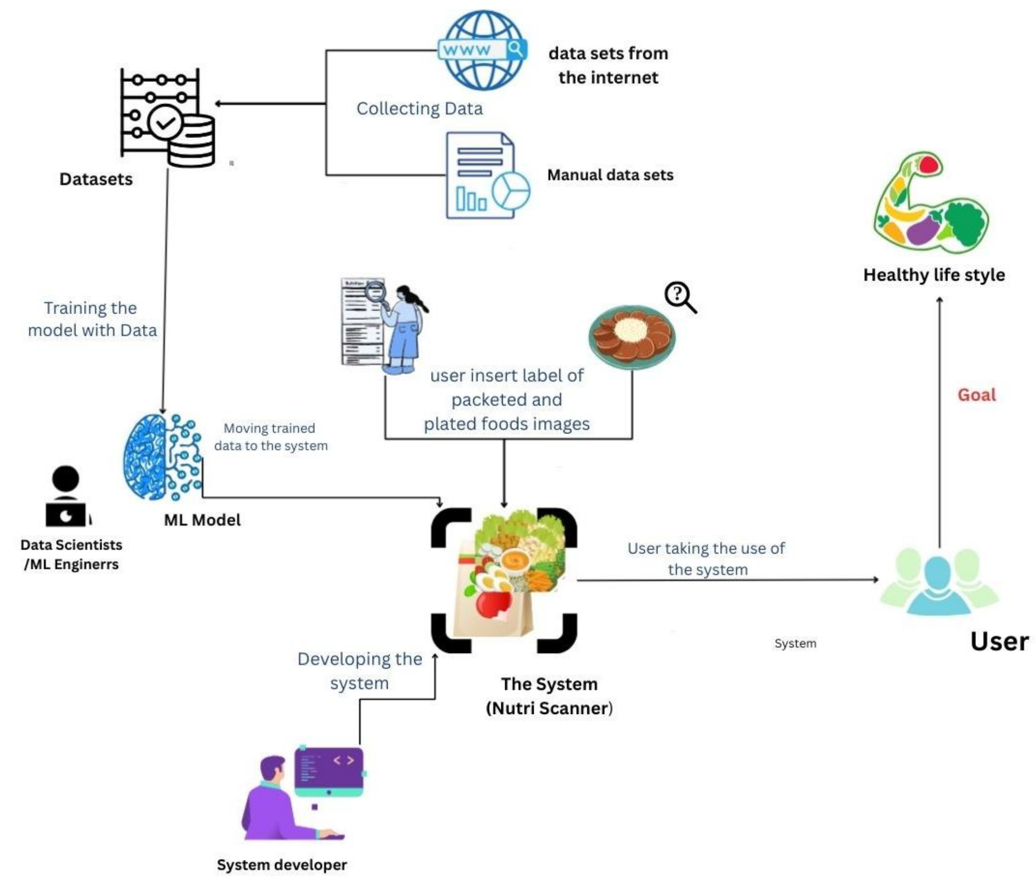
1. Introduction

This chapter provides an overview of how the system requirements for the *Nutri Scanner* project were gathered and analyzed. It begins by identifying the key stakeholders and their roles, followed by a discussion of the methods used to collect and evaluate requirements such as surveys, interviews, and literature reviews. The chapter also presents diagrams, like use case and activity diagrams, to illustrate system interactions. Finally, it defines the project's scope by outlining the functional and non-functional requirements, including features like food image recognition, multilingual OCR, and health risk prediction, ensuring all requirements are prioritized to guide the system's design and development effectively.

1.1. Rich picture

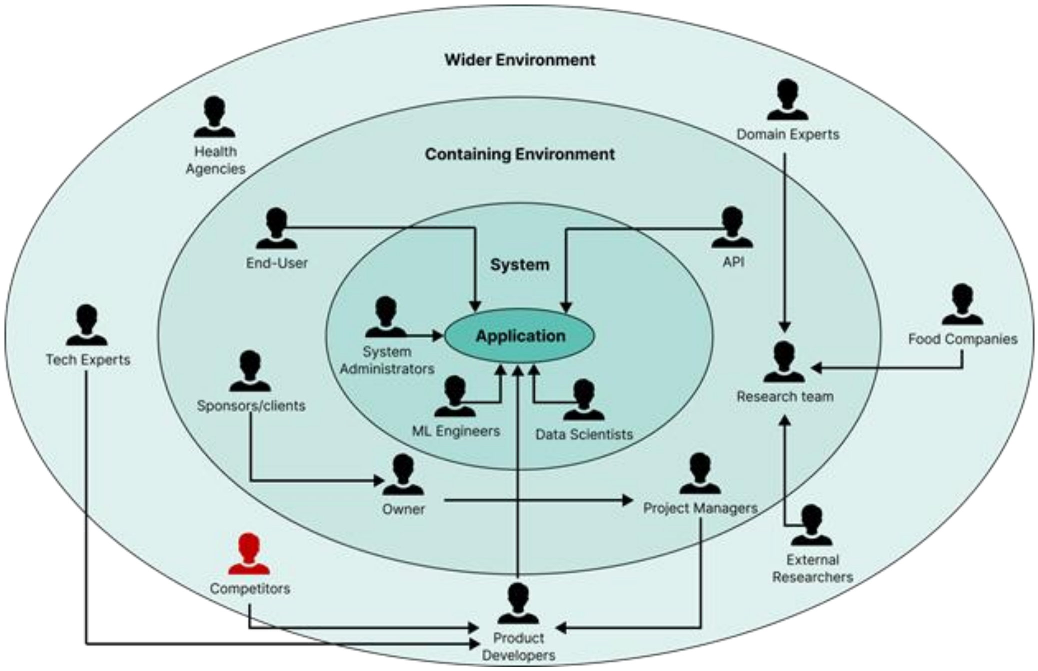


1.3 Stakeholder Analysis

Stakeholder Viewpoints

The Onion model below displays the Stakeholders that are linked to the system. To gain an understanding of each Stakeholder's position, it is represented as it is depicted in the figure below.

1.3.1. Onion Model



Stakeholder	Role	Benefit
ML engineer Data Scientist	Maintains Operations	Develops and manages an application providing support to datasets, ontologies, and functional APIs
System Administrator	Administrates Operations	manages and maintains the application's infrastructure
End-User	Uses the Application	Directly uses the System to get nutritional information
Sponsor/Client	Provides Financial Aid	Funds the project and operations
Owner	Functional Beneficiary	stakeholder of the overall vision, success, and long-term direction
Project Manager	Manages the Application	Oversee aspects of the project such as timelines, tasks, and coordination
Research Team	Gathers Information	Conducts Studies to aid in the development of the project by gathering tools and data.
API	Functional Beneficiary	An External Software interface that allows the system to connect and exchange data with other applications
Domain Experts	Industry Expert	Specialists in the nutritional Sector providing guidance
Food Companies	Industry Expert	Businesses that provide sample data
External Researchers	3rd Party Researcher	Independent researchers outside of the development team who provide the necessary data or tools
Product Developers	Develops the System	Team responsible for developing and maintaining the system
Competitors	Negative Stakeholder	Competes against the proposed System
Tech Experts	Industry Expert	Specialists in the technology sector provide guidance
Health Agencies	Regulatory agency	Regulatory government bodies that ensure guidelines

1.4 Selection of Requirement Elicitation Techniques/Methods

1.4.1 Observing Current Systems and Literature Review (LR)

A literature study of existing nutrition tracking systems and research is essential for eliciting requirements. This analysis reveals characteristics and gaps **in nutrition analysis, OCR-based food recognition, and personalized diet recommendation systems**.

Advantages	Disadvantages
Explores key components and AI algorithms like **computer vision** for food recognition, OCR for label detection, and health risk models	Academic/medical research utilizes complex terminology that **requires domain expertise**.
Identifies global datasets, tools, and nutritional standards (e.g., USDA, WHO, FAO)	Many global solutions may **not fit the Sri Lankan context** due to different diets .
Reveals feature gaps such as a **lack of bilingual** (Sinhala-English) food label analysis and **traditional food recognition** in current systems.	Requires local interpretation and **cultural adaptation** of research findings.

1.4.2 Surveys and Questionnaires

A survey is suitable for Nutri Scanner to efficiently gather requirements from diverse users, including those with chronic conditions and health-conscious individuals

Advantages	Disadvantages
Reaches a **diverse audien**ce, including patients, healthcare professionals, and users.	Respondents may misinterpret questions, leading to **incomplete or inaccurate data.**
Cost-effective and time-efficient for collecting broad data on eating habits, nutritional awareness, and user needs.	Not all responses will follow the required format, causing **inconsistent results**.
Helps identify **common dietary issues** and user expectations for AI-based nutrition tools.	Limited ability to gain **insights** into personal motivation behind food choices and lifestyle.

1.4.3 Interviews

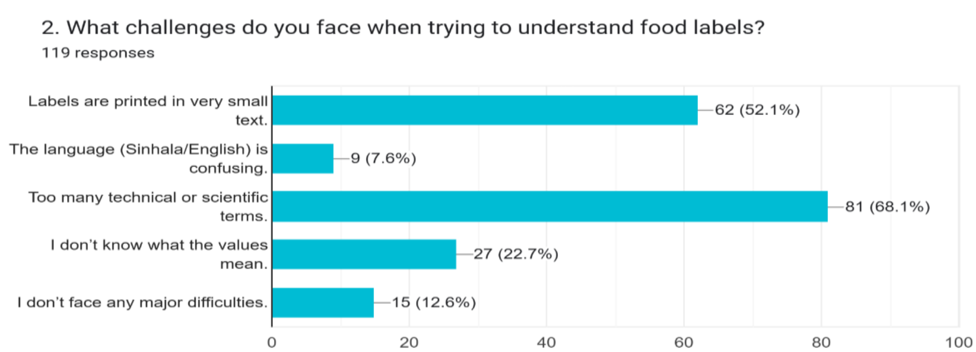
Interviews with nutritionists, healthcare professionals, and users provide deeper insights into needs and challenges in nutrition tracking and health risk prediction.

Advantages	Disadvantages
Direct interaction allows clarification of doubts and detailed feedback.	**Time-consuming** and limits number of participants.
Provides **qualitative insights** into real user experiences and pain points	Responses may be **subjective and vary** by personal opinion.
Helps gather **expert advice** on nutrition and health management	Requires **skilled interviewers** for accurate interpretation.

1.5 Discussion of results

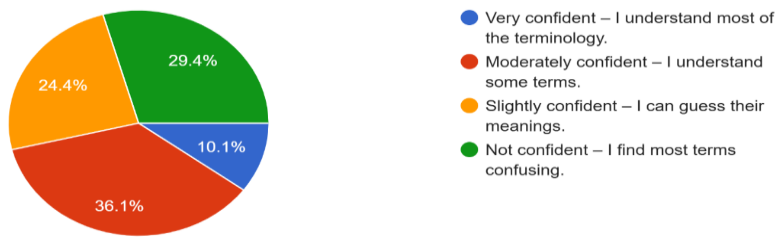
Question	How often do you actually read the nutrition label before eating something?
Aim of Question	The aim is to understand how users notice and use nutrition labels on packaging, revealing whether they pay attention or ignore this information. This insight guides the need for automated tools like NutriScanner to help interpret nutrition labels effectively.
Observations	

Conclusion	Most respondents (about 52%) only read nutrition labels occasionally, and 21.8% rarely, with only 20.2% always reading labels. The main reasons for limited reading are small text and complex language, emphasizing the need for a simple, AI-driven system like Nutri Scanner to make nutritional data more accessible and understandable.
Question	What challenges do you face when trying to understand food labels?
Question aim	Identify user challenges in reading food labels, language, and nutrition understanding
Observations	



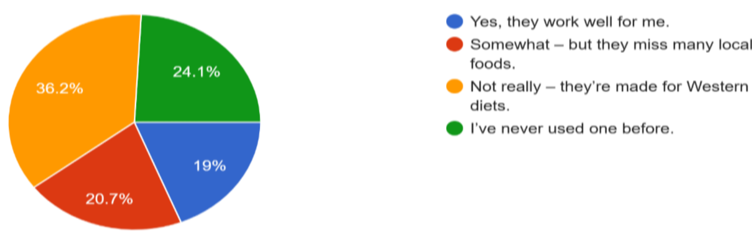
Conclusion	Most participants face challenges interpreting food labels: 68.1% struggle with technical terms, 52.1% with small print and mixed Sinhala-English text, 22.7% do not fully understand nutritional values, and only 12.6% have no major issues. These barriers highlight poor label readability, language complexity, and limited nutrition knowledge, underscoring the need for NutriScanner's AI to simplify labels for Sri Lankan users.
Question	How confident are you in understanding terms commonly found on food labels (e.g., saturated fat, sodium)?
Aim of Question	To evaluate user confidence in understanding key nutritional terms on food labels and their familiarity with common dietary components like fats, sugars, and sodium.
Observations	

3. How confident are you in understanding terms commonly found on food labels (e.g., saturated fat, sodium)?
119 responses



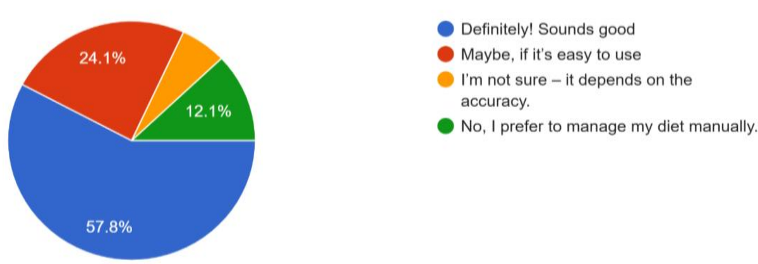
Conclusion	Most users lack confidence in understanding nutritional terms: only 10.1% are very confident, 36.1% moderately confident, 24.4% slightly confident, and 29.4% not confident or unfamiliar with terms like saturated fat and sodium. This highlights the need for Nutri Scanner to simplify and localize these terms to help users make informed dietary choices.
Question	Do you think existing mobile nutrition or calorie-tracking apps fit Sri Lankan diets and food culture?
Aim of Question	If current nutrition and calorie-tracking apps support Sri Lankan dietary culture and local foods, or if gaps exist in representing local meal plates and packaged items.
Observations	

4. Do you think existing mobile nutrition or calorie-tracking apps fit Sri Lankan diets and food culture?
116 responses

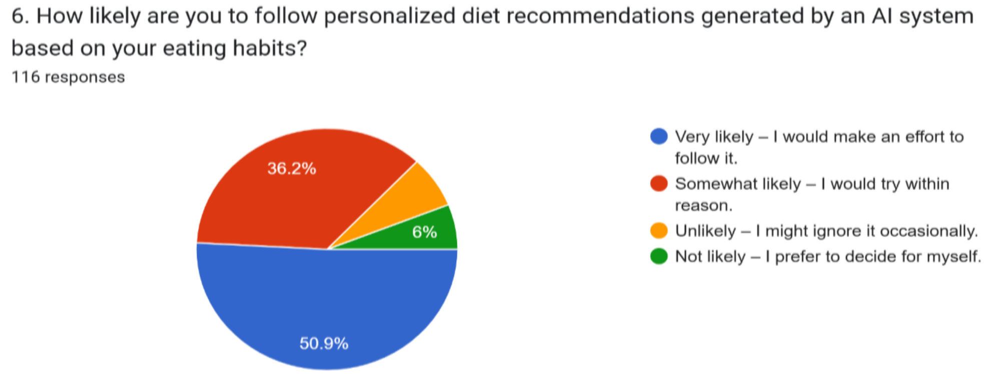


Conclusion	Most participants feel current apps insufficiently support Sri Lankan food culture, with 36.2% saying apps focus on Western diets and 20.7% noting a lack of local food options. This gap highlights the need for a specialized solution like NutriScanner tailored to local needs.
Question	Would you use an app that scans your food and tells you if it's healthy (Sri Lankan style)?
Aim of Question	To gauge potential user interest and adoption of a localized food scanning app targeting Sri Lankan cuisine.
Observations	

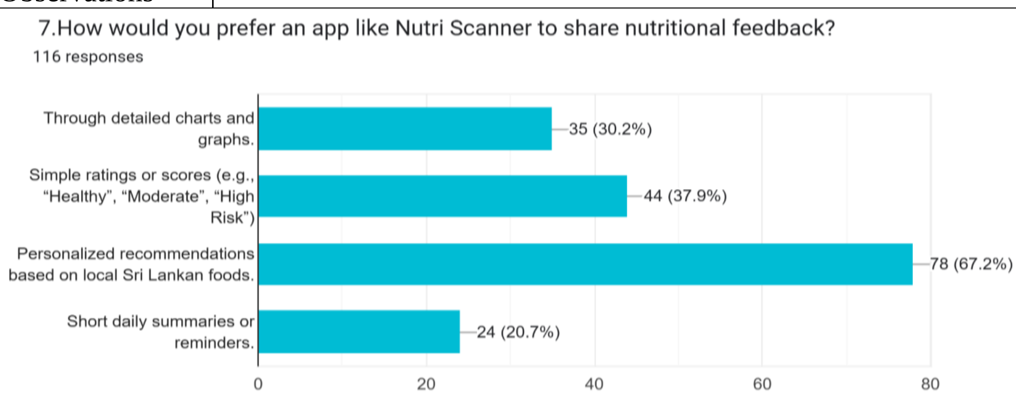
5. Would you use an app that scans your food and tells you if it's healthy (Sri Lankan style)?
116 responses



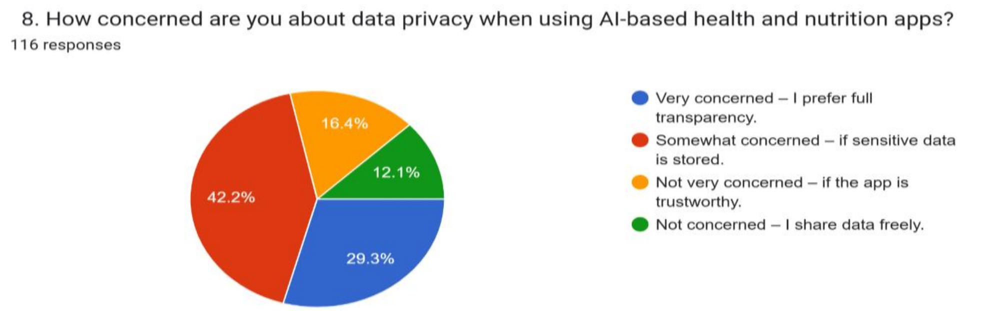
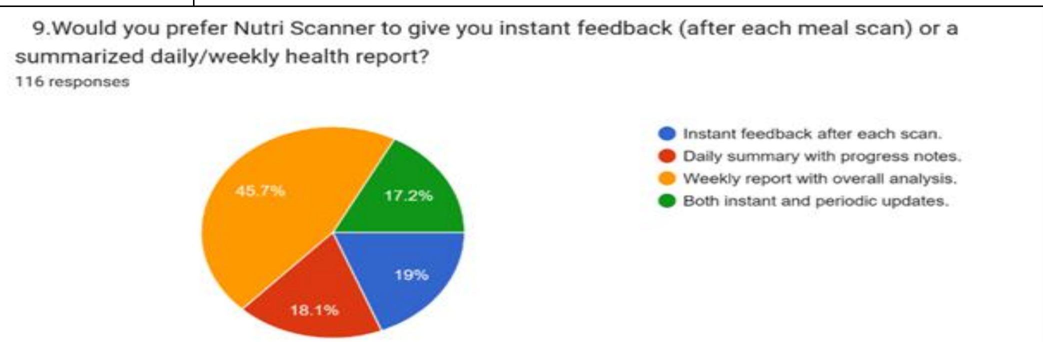
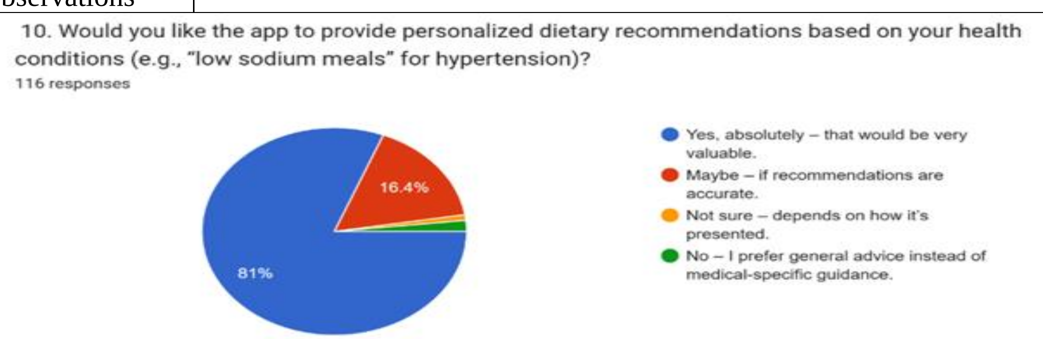
Conclusion	A clear majority (57.8%) indicated strong interest in a Sri Lankan-style healthy food scanning app. The high proportion of positive responses underscores user enthusiasm for digital nutrition support, while the importance of usability and accuracy remains a key factor that could further boost acceptance.
Question	How likely are you to follow personalized diet recommendations generated by an AI system baes on your eating habits?
Aim of Question	To assess the willingness of users to adopt and follow dietary recommendations generated by AI systems based on individualized data.
Observations	



Conclusion	Over 50% of respondents are willing to follow personalized diet recommendations, indicating strong trust in AI-driven, adaptive nutrition advice like NutriScanner.
Question	How would you prefer an app like Nutri Scanner to share nutritional feedback?
Aim of Question	To determine users' preferred mode of receiving nutritional insights and feedback from a nutrition app, especially considering local content and personalization.
Observations	



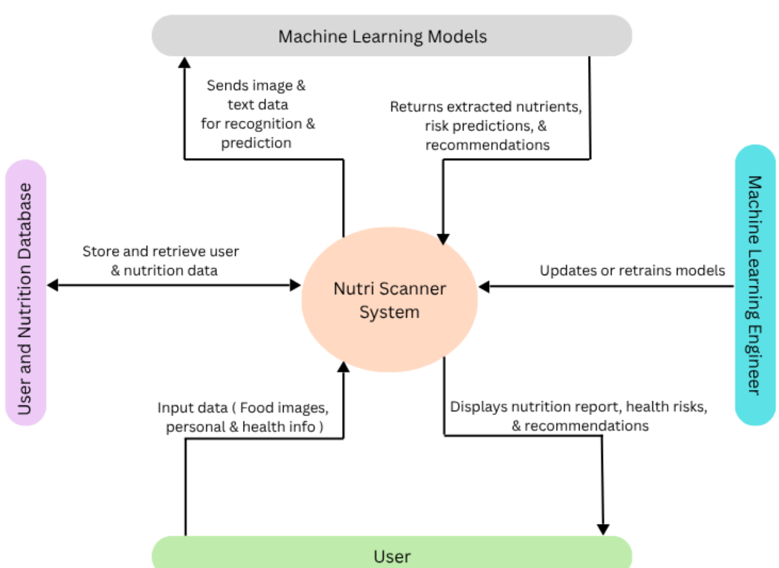
Conclusion	A majority (67.2%) prefer personalized nutrition advice based on local Sri Lankan foods, while others favor simple ratings (37.9%) or detailed charts (30.2%), with fewer liking short daily summaries (20.7%). This shows strong demand for culturally tailored feedback, supporting NutriScanner's focus on local personalization.
Question	How concerned are you about data privacy when using AI-based health and nutrition apps?
Aim of Question	To assess user attitudes and concerns regarding data privacy when engaging with AI-driven nutrition and health applications.
Observations	

	
Conclusion	Most respondents are concerned about data privacy: 29.3% are very concerned, and 42.2% are somewhat concerned about sensitive data storage. Only 16.4% are less worried if the app is trustworthy, and 12.1% are unconcerned, highlighting the need for strong privacy, transparency, and data protection in NutriScanner.
Question	Would you prefer Nutri Scanner to give you instant feedback (after each meal scan) or summarized daily/weekly health report?
Aim of the question	To identify user preferences regarding feedback frequency and format. This helps determine the optimal structure for delivering insights and maintaining engagement.
Observations	
	
Conclusion	Users show mixed feedback preferences for NutriScanner: 45.7% prefer weekly comprehensive reports, 19% want instant feedback, and 17.2% like both options. This highlights the need to offer flexible feedback modes to suit different user preferences.
Question	Would you like the app to provide dietary recommendations based on your health conditions
Aim of the question	To assess user interest in personalized dietary guidance tailored to their health needs, determining demand for a medically adaptive nutrition system in Nutri Scanner.
Observations	
	
Conclusion	A large majority (81%) expressed strong interest in personalized dietary recommendations, 16.4% were unsure pending advice accuracy, and only a few preferred general suggestions. This shows clear demand for individualized nutrition insights.

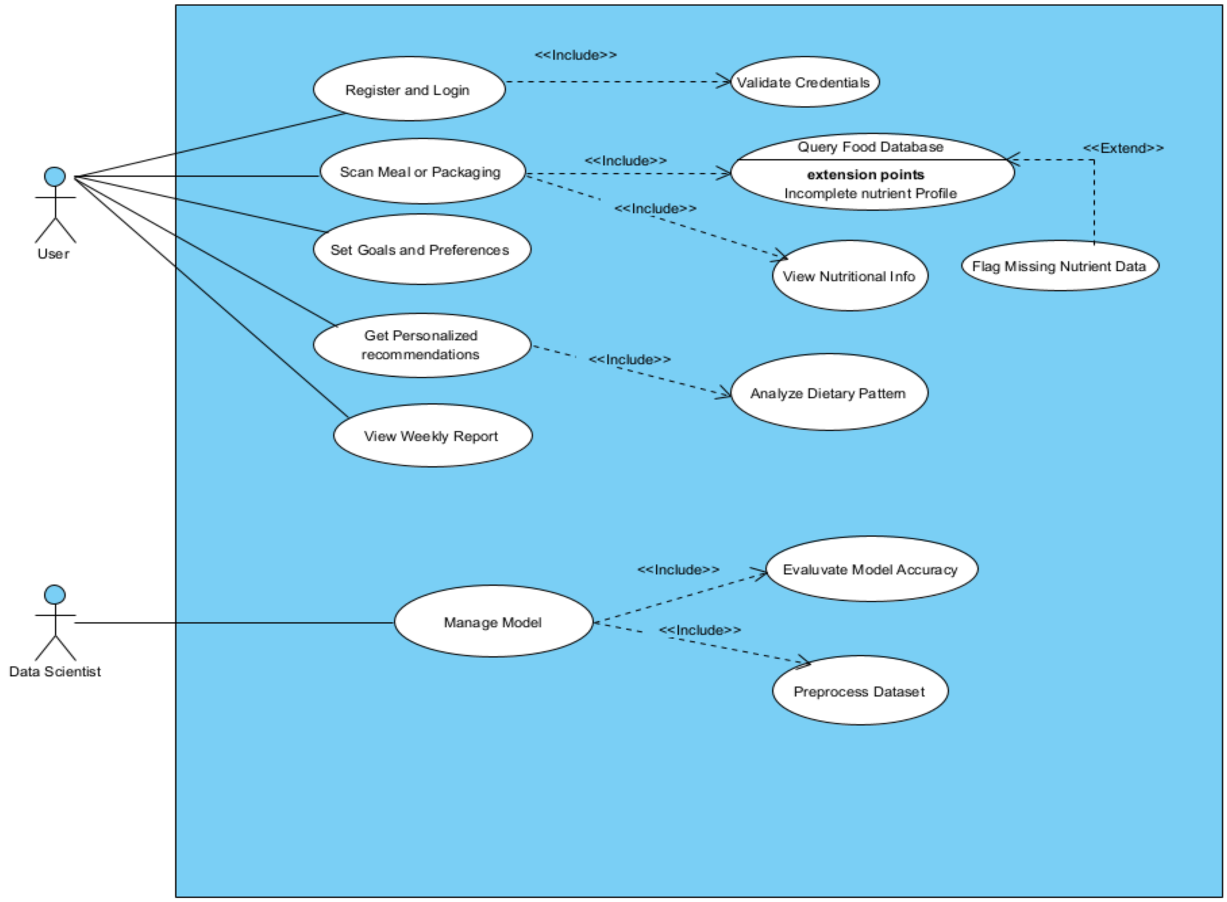
1.6. Summary of Findings

Findings	LR	Questionnaire	Existing Systems
Users rarely read nutrition labels due to small font size, language confusion, and complex terminology.	X	X	
Many individuals lack confidence in interpreting nutritional terms like “saturated fat” or “sodium.”	X	X	
Existing apps and tools are not fully adapted to Sri Lankan diets and local food culture.	X	X	X
Personalized diet recommendations generated by AI are generally welcomed, but users expect accuracy and local relevance.	X	X	X
Users prefer simple, visual, and personalized feedback rather than complex numeric data.	X	X	
Current apps lack integration of local foods and fail to generate relevant meal suggestions for Sri Lankan users.	X		X
Users are concerned about data privacy and transparency in AI-based nutrition apps.	X	X	
Most existing apps do not provide condition-specific dietary recommendations (e.g., low sodium for hypertension).	X	X	X
Overall, there is a strong need for a unified system that offers scanning, analysis, feedback, and personalized recommendations for local diets.	X	X	X

1.7 Context Diagram



1.8. Use Case Diagram



1.8.1Use Case Descriptions

Use Case ID	**UC001**
Use Case Name	Register and Login
Description	User creates an account and logs into the Nutri Scanner system to access personalized nutrition tracking features
Participating Actors	User
Pre-conditions	● User has a valid email address or phone number
Main Flow	1. User navigates to the registration page 2. User enters required credentials (email, password, personal information) 3. System validates the entered credentials 4. System creates user account and sends confirmation 5. User logs in with credentials 6. System authenticates user and grants access to the application
Alternative Flow	If the user already has an account, they can directly proceed to login without registration
Exceptional Flows	E1. Invalid credentials provided - system displays error message E2. Email already exists - system prompts user to log in

Use Case ID	**UC002**
Use Case Name	Scan Meal or Packaging
Description	User scans food items or packaging to retrieve nutritional information.
Participating Actors	User
Pre-conditions	• User is logged into the system • Camera permissions are granted • Food database is accessible
Main Flow	1. User selects the scan option from the main menu 2. User points the camera at a meal or food packaging 3. The system captures an image and processes it 4. System queries the food database for nutritional information 5. System displays nutritional profile to the user
Alternative Flow	If the food item is not found, the system prompts the user to query the database manually or flag missing data
Exceptional Flows	E1. Image quality is poor - the system requests the user to rescan E2. Food item not found in database - system displays incomplete nutrient profile message

Use Case ID	**UC003**
Use Case Name	Set Goals and Preferences
Description	User defines personal nutritional goals, dietary preferences, and restrictions for personalized recommendations
Participating Actors	User
Pre-conditions	• User is logged into the system • User profile exists
Main Flow	1. User navigates to goals and preferences settings 2. User inputs dietary goals (weight loss, muscle gain, maintenance, etc.) 3. User specifies dietary preferences (vegan, vegetarian, keto, etc.) 4. User sets nutritional targets (calorie limit, protein goal, etc.) 5. System saves goals and preferences
Alternative Flow	User can skip setting goals initially and configure them later
Exceptional Flows	E1. Invalid goal parameters entered - system displays validation error

Use Case ID	**UC004**
Use Case Name	Get Personalized Recommendations
Description	System analyzes user's dietary pattern and provides personalized meal and nutrition recommendations
Participating Actors	User
Pre-conditions	• User has set goals and preferences • Sufficient dietary data is available
Main Flow	1. User requests personalized recommendations 2. System analyzes dietary patterns from scanned meals 3. System compares current intake with user goals

	4. System generates personalized meal recommendations 5. System displays recommendations to user
Alternative Flow	If insufficient data is available, system provides general recommendations based on goals
Exceptional Flows	E1. Analysis fails due to insufficient data - system notifies user to scan more meals

Use Case ID	**UC005**
Use Case Name	View Weekly Report
Description	User views a comprehensive weekly summary of nutritional intake, progress toward goals, and insights
Participating Actors	User
Pre-conditions	• User has logged meals during the week
Main Flow	1. User navigates to the weekly report section 2. System retrieves user's meal data for the past week 3. System calculates nutritional totals and averages 4. System generates visual charts and progress indicators 5. System displays comprehensive weekly report
Alternative Flow	User can select custom date ranges for viewing reports
Exceptional Flows	E1. No data available for the period - system displays message E2. Report generation fails - system displays error and suggests retry

Use Case ID	**UC006**
Use Case Name	Manage Model
Description	Data scientist manages the complete machine learning pipeline including uploading training data, training models, evaluating accuracy, and deploying models to production.
Participating Actors	Data Scientist
Pre-conditions	• Access to model development platform • Valid datasets are prepared • Required computational resources are available
Main Flow	1. Data scientist uploads and organizes training datasets 2. Data scientist preprocesses and cleans the data 3. Data scientist selects model architecture and configures hyperparameters 4. Data scientist initiates model training process 5. System trains model using prepared datasets 6. Data scientist evaluates model accuracy using test data 7. Data scientist reviews performance metrics and makes adjustments if needed 8. Data scientist deploys the trained model to production environment 9. System confirms successful deployment

Alternative Flow	If model accuracy is unsatisfactory, data scientist can retrain with adjusted parameters or additional data
Exceptional Flows	E1. Invalid dataset format - system displays format error E2. Training fails due to insufficient resources - system displays error E3. Model performance below threshold - system flags for retraining E4. Deployment fails - system rolls back and alerts data scientist

9. Functional Requirements (with prioritization)

The system has different levels, which are defined with different priorities. This section shows that information here.

Priority Level	Description
Critical	Main possible goals or objectives of the system
Important	Not mandatory, might be needed
Non-important	Requirements which are out of scope

	Requirement and Description	Priority
FR01	Accurately recognize and extract bilingual nutrition labels (Sinhala and English) on packaged foods	Critical
FR02	Identify traditional homemade Sri Lankan dishes from user-uploaded images and estimate nutrition	Critical
FR03	Normalize nutritional information using local and USDA databases	Critical
FR04	Allow users to input personal health data (age, gender, weight, medical conditions)	Important
FR05	Predict health risks (e.g., diabetes, hypertension) based on nutrition intake and personal health data	Critical
FR06	Provide personalized dietary recommendations and alerts based on health risk and nutrient intake	Critical
FR07	Offer an intuitive user interface for image uploads, nutrition visualization, and recommendations	Important
FR08	Securely store and manage user data with privacy compliance	Critical
FR09	Regularly update food databases and models with new data and feedback	Non-important
FR10	Support export or sharing of nutritional reports	Non-important

1.10. Non-Functional Requirements

• **Accuracy:** Very important for correct recognition and interpretation of nutrition labels or food images, which minimizes errors in the extraction and analysis of nutritional information.

• **Performance:** Crucial in offering speedy responses and timely nutrition analysis for a seamless user experience, under normal operation loads**.**

• **Security:** This is key to protecting information from users, encrypting data in storage and transmission, ensuring privacy and security against unauthorized access.

• **Maintainability:** the possibility of improvement in the future for ease of updating of fodatabases and system components, which will contribute to long-term upgradability and sustainability.

• **Data Privacy Compliance**: Imperative to guarantee consent from users and their rights related to data collection and deletion for trust-building and meeting ethical standards.

1.10.1 Requirements and Description

Non-Functional Requirement (NFR)	Specification	Priority
NFR1: Accuracy	The OCR and food recognition modules shall achieve at least 90% accuracy in text and image interpretation.	Critical
NFR2: Performance	The system shall respond to user inputs and provide nutrition analysis within 5 seconds under normal load.	Critical
NFR3: Usability	The app shall have an intuitive and simple interface, ensuring that users with varying tech literacy can easily use the system.	Important
NFR4: Security	The system shall encrypt all user data to protect privacy and ensure secure storage and transmission.	Critical
NFR5: Scalability	The system shall support increasing numbers of users and growing food datasets without performance degradation.	Non important
NFR6: Reliability and Availability	The system shall have 99.5% uptime and gracefully recover from failures with minimal user impact.	Non important
NFR7: Multilingual Support	The system shall fully support Sinhala and English for UI	Important

NFR8: Compatibility	The app shall be compatible with Android and iOS mobile platforms, supporting the last two major OS versions.	Non Important
NFR9: Maintainability	The software shall be modular and documented to allow for easy updates to food databases and models.	Non Important
NFR10: Data Privacy Compliance	The system shall ensure user consent for data collection and provide mechanisms for data deletion upon request.	Critical

1.11. Chapter summary

This chapter summarizes the process of identifying stakeholders and understanding their influence on the system requirements for the Nutri Scanner project. The mechanism of eliciting the requirements from target users, such as local consumers and health-conscious individuals, is explained through requirement elicitation methods used during this research work, including literature reviews, surveys, and interviews. This chapter also provides an overview of the major functional and non-functional requirements driving the system, particularly in the areas of food image recognition, multilingual OCR processing, nutrient estimation, and personalized health risk prediction. These analyses helped determine the priority of the requirements needed to achieve the project's overall goal of delivering an accurate, user-friendly, and locally relevant nutrition tracking solution.